SWEN30006
Software Modelling and Design

Domain Models & System Sequence Diagrams

Textbook: Larman Chapter 1, 9 & 10

"It's all well in practice, but it will never work in theory."

— anonymous management maxim

SWEN30006 - Lecture Questions

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Nobody has responded yet.

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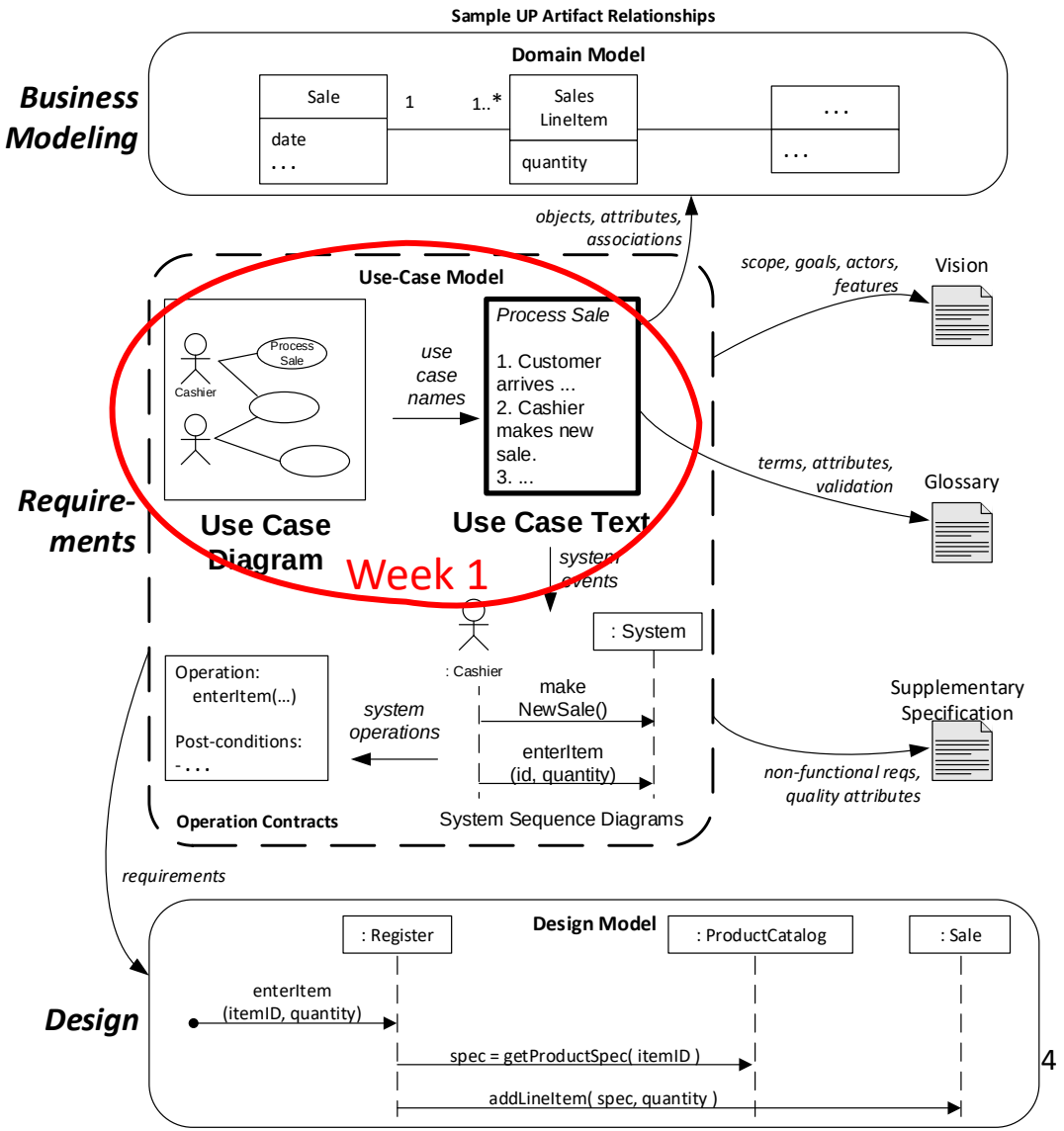
Learning Objectives

On completion of this topic, you should be able to:

- State the purpose of OO Analysis.
- Be familiar with static and dynamic OO models
  - Domain Class Diagrams
    - Identify conceptual classes
    - Create an initial domain model
    - Model appropriate attributes and associations
  - System Sequence Diagrams (SSD)
    - Identify system events
    - Create SSDs for use case scenarios

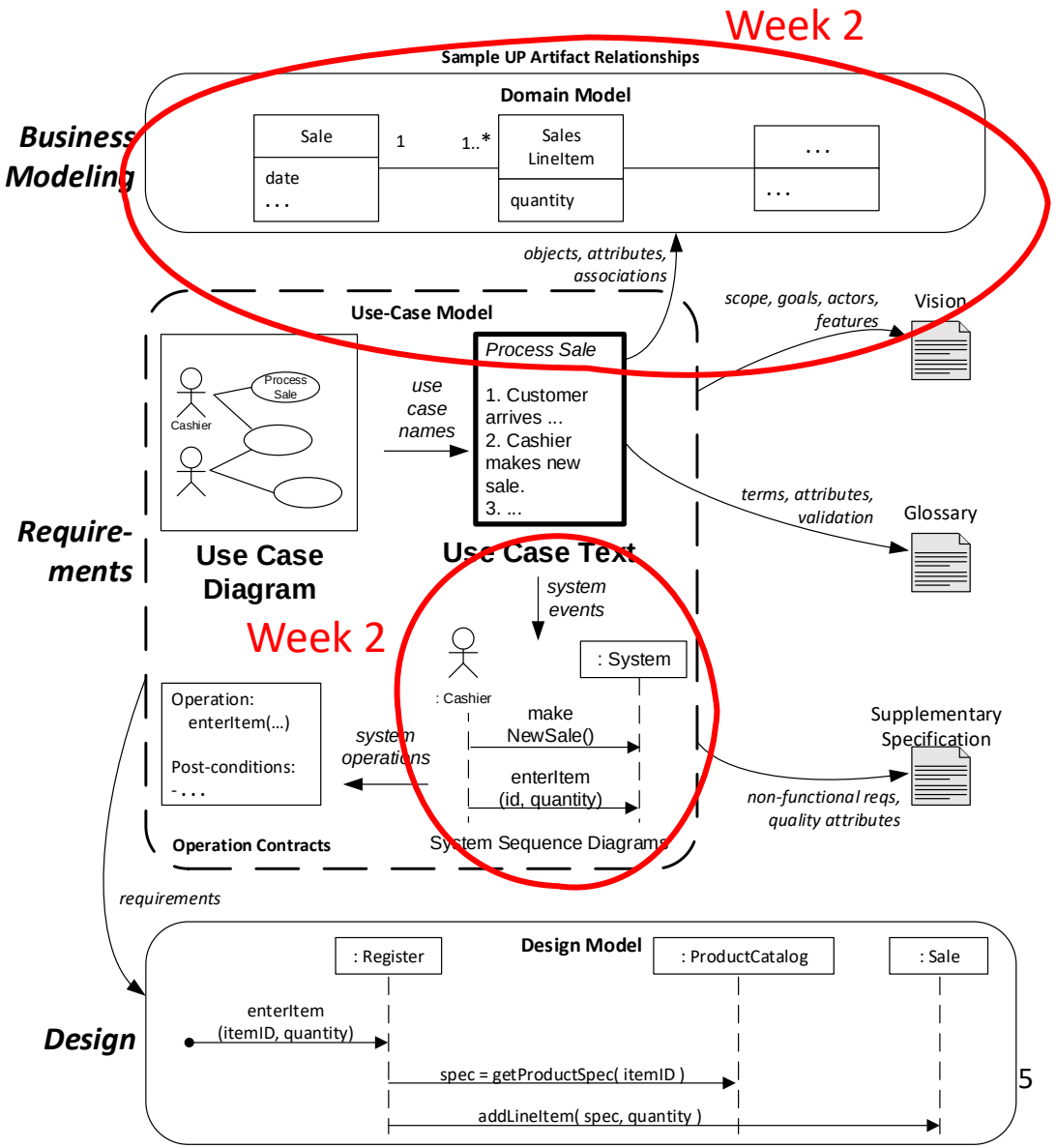
Week 1 Revisited: Use Cases

- **Use Cases** are text stories that describe how an actor *use* a system to meet goals
  - A tool to discover and record requirements
- **Use case diagrams** provides a context of a system and its environment
  - A tool to capture the names of use cases, actors, and their relationships



Week 2: Object-Oriented Analysis

- Week 2 focuses on object-oriented analysis based on use case text
- *Analysis* is investigation of the problem & requirements
- ***Object-oriented analysis*** is a process of creating a description of the domain from the perspective of *object*.
  - Analyse use cases and identify objects – or concepts – in the problem domain
  - The concepts and behaviours are captured in the form of *Domain Models* and *Sequence Diagrams*



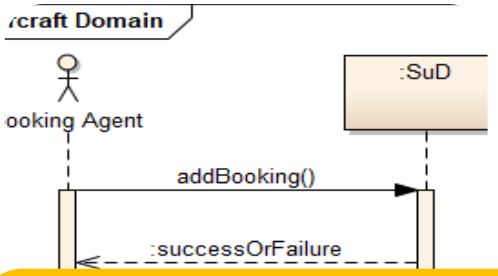
OO Analysis, Design, and Development

Problem domain

- Concepts
- Relationships

Use Cases

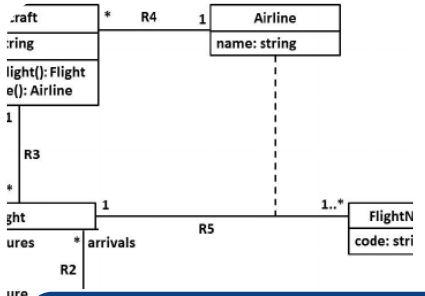
Text stories



OO Domain Models

- Static: Domain Class Diagram
- Dynamic: System Sequence Diagram

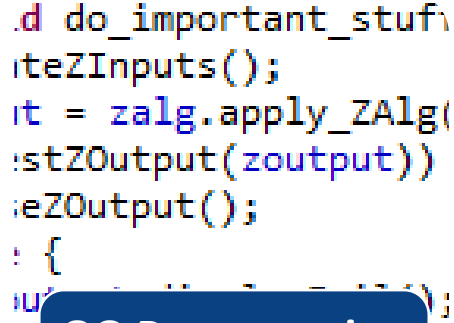
Conceptual Classes



OO Design Models

- Static
- Dynamic

Software Classes

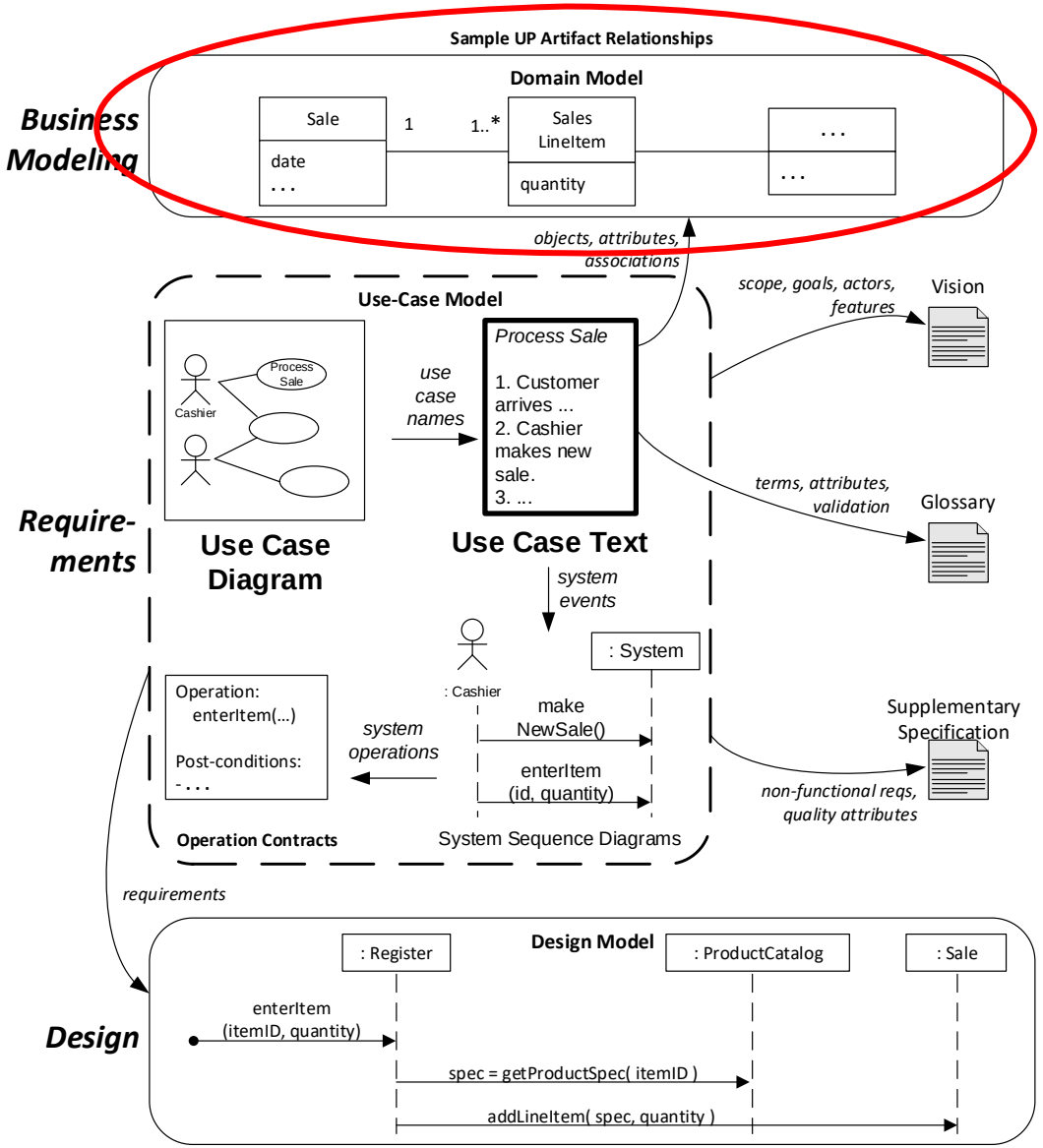


OO Programming

Classes in Programming

Domain Models

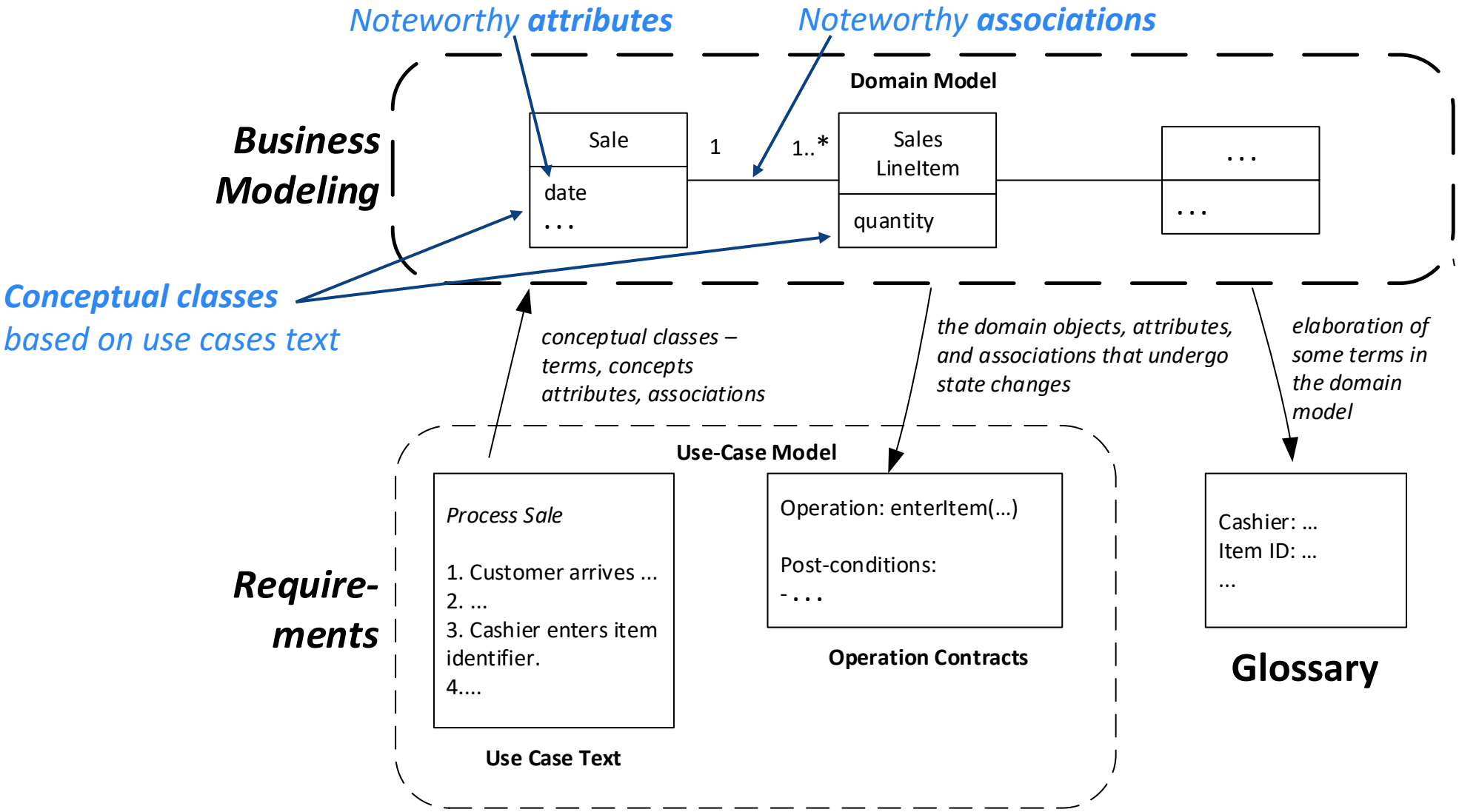
Textbook: Larman Chapter 9



OO Analysis & Domain Models

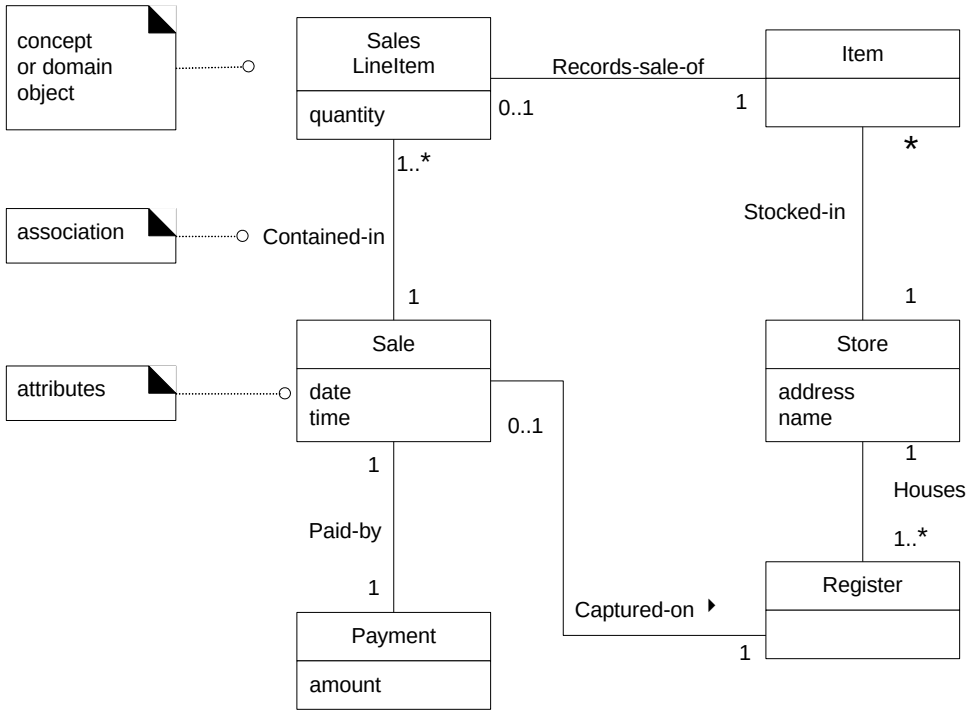
- OO Analysis includes an identification of the **concepts**, **attributes**, and **associations** that are considered noteworthy in the problem domain – These can be expressed by *a domain model*
- **Definition:** A domain model is a representation of *real-situation* conceptual classes
  - *Not* a software object
  - Shows the *noteworthy* domain concepts or object
  - One of OO artifacts
- The domain model = UP (Unified Process) Business Object Model
  - Focus on explaining *things* and products important to a business domain
- **Visual** representation: UML class diagram is used to illustrate a domain model
  - A domain class diagram provides a *conceptual* perspective to show conceptual classes, their attributes and associations with other classes
  - *No* operations (method signatures) are defined in the class diagram

Domain Models in UP

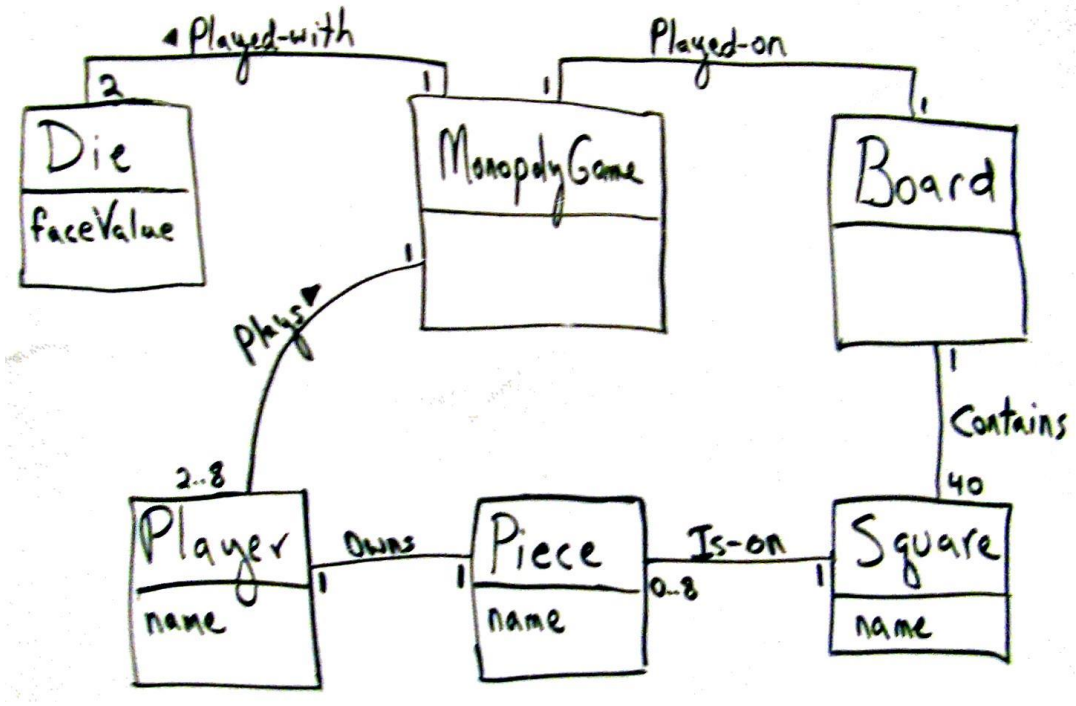


Domain Model: A Visual Dictionary

- The domain model is a *visual dictionary* of the noteworthy abstractions, domain vocabulary, and information content of the domain



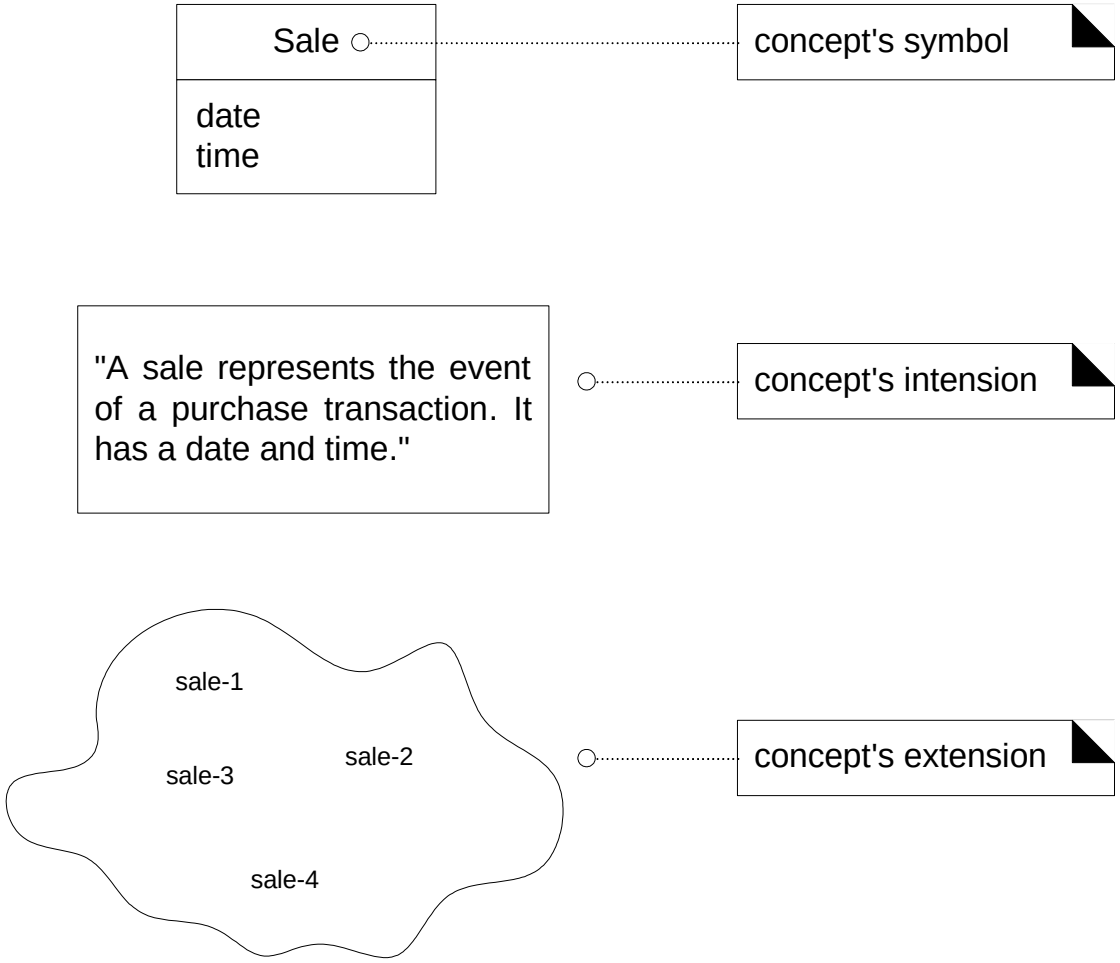
A partial domain model for POS



A partial domain model for Monopoly

A Conceptual Class: Definition

- **(Informal) Definition:** A conceptual class is an idea, thing, or object
  - It can be more than a data model. It can have a purely behavioural role in the domain instead of an information role
- **(More formal) Definition:** A conceptual class may be considered in terms of its symbol, intension, and extension
  - **Symbol:** Words or images representing a conceptual class.
  - **Intension:** The definition of a conceptual class.
  - **Extension:** The set of examples to which the conceptual class applies.



A Conceptual Class: How to identify it?

Three strategies to find a conceptual class from use cases

1. Identify based on the existing models for common domain problems, e.g., inventory, finance, etc.
   - Books: *Analysis Patterns by* Martin Fowler, *Data Model Patterns* by David Hay
2. Use a category list
   - See the conceptual class category list in Table 9.1 in the textbook (Larman Chapter 9)
3. Identify noun phrases
   - Care must be applied as words in natural languages are ambiguous
   - Do not use as a mechanical noun-to-class mapping

Conceptual Class Category	Examples
Business transactions Guideline: Critical (involve money)	Sale, Payment Reservation
Physical objects Guideline: esp. for device-controllers or simulations	Item, Register Board, Piece, Die Aeroplane
Containers of things (physical or information)	Store, Bin Board Aeroplane
Where are transactions recorded? Guideline: important	Register, Ledger FlightManifest, MaintenanceLog

Example: POS

Main Success Scenario (or Basic Flow):

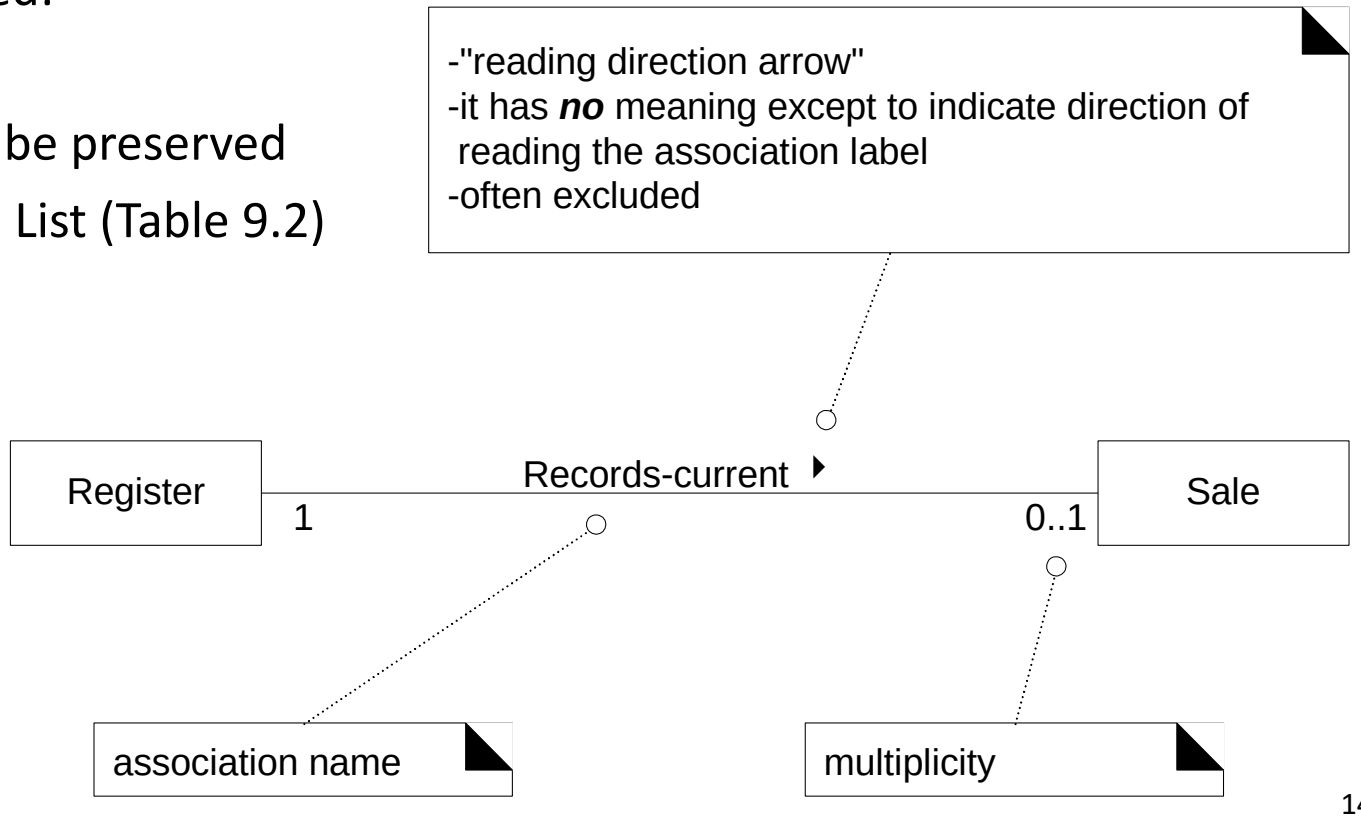
1. Customer arrives at a POS checkout with goods and/or services to purchase.
2. Cashier starts a new sale.
3. Cashier enters item identifier.
4. System records sale line item and presents item description, price, and running total. Price calculated from a set of price rules.

Identify conceptual classes based on a category list

- Role of people or organization:
  - Customer, Cashier
- Product or services:
  - Goods and/or services (rename to item)
- Business transaction:
  - Sale
- Transaction line items:
  - Sale line item
- Description of things:
  - Item description
- Where is the transaction recorded?
  - A POS checkout (Register)

Associations between Conceptual Classes

- **Definition:** An *association* is a relationship between classes that indicate some meaningful and interesting connection
- The following associations should be included:
  - Significant in the domain
  - Knowledge of the relationship needs to be preserved
  - Derived from the Common Associations List (Table 9.2)



*See Workshop 2 supplementary materials

Identify Conceptual Classes (Click Potential Terms on Image)

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- The library has users who can borrow items.
- The library holds both books and videos: all items have a title and year of production (and a due date if they have been borrowed), while only books have authors and only videos have a duration.
- There are different kinds of users: students, staff, and guests. The only difference between these user types is that they have a different borrowing limit (max number of items they can borrow) which is based on their user type.
- The library keeps track not only of which items are out on loan now, but the complete history of which users borrowed which items for which periods.

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Example: Conceptual classes

- The library has **users** who can borrow **items**.
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Similar Concepts

- In the problem domain, there might be concepts that look very similar

The library has **users** who can borrow **items**.

The library holds both **books** and **videos**: all **items** have a *title* and *year* of production (and a due date if they have been borrowed), while only books have *authors* and only *videos* have a duration

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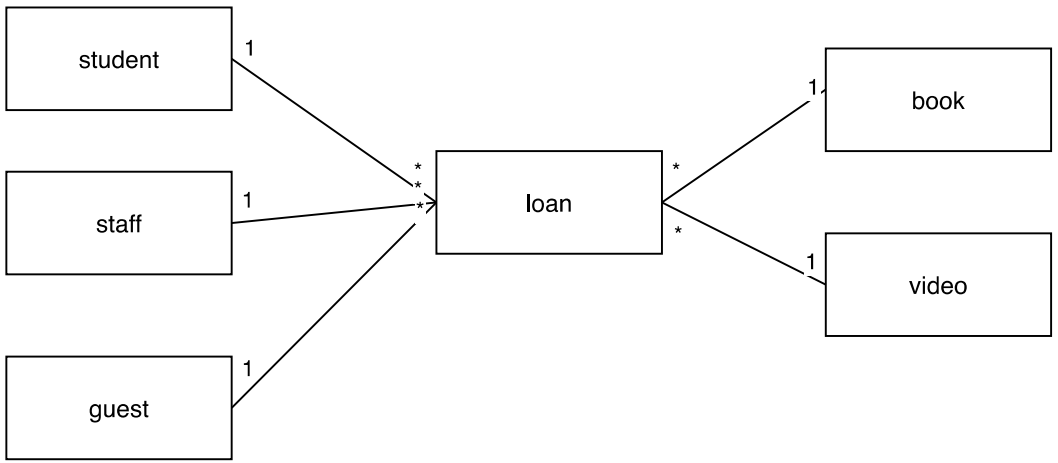
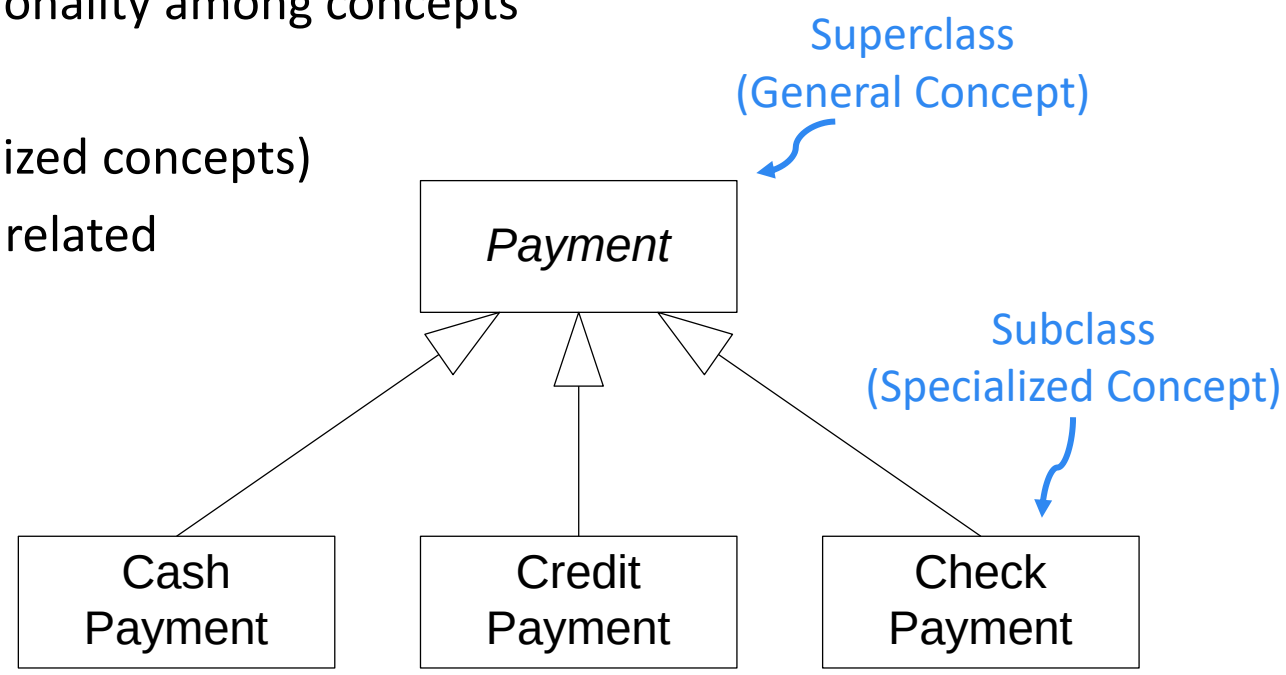


Partial domain model

- The conceptual classes are not well organized.
- The association between items, books, and video is missing. Same for users

Domain Model: Generalization-Specialization Class Hierarchy (or Class Hierarchy)

- When concepts are very similar, it is valuable to organise them
- ***Generalization:*** The activity of identifying commonality among concepts
  - Define *superclass* (general concept)
  - Define relationships with *subclasses* (specialized concepts)
  - Conceptual subclasses and superclasses are related in terms of set membership
- *Why?*
  - Economy of expression
  - Reduction in repeated information
  - Improved comprehension



A Domain Class Model

See more in Textbook Chapter 31

Domain Model with Class Hierarchy

- Organise the similar concepts as subclasses

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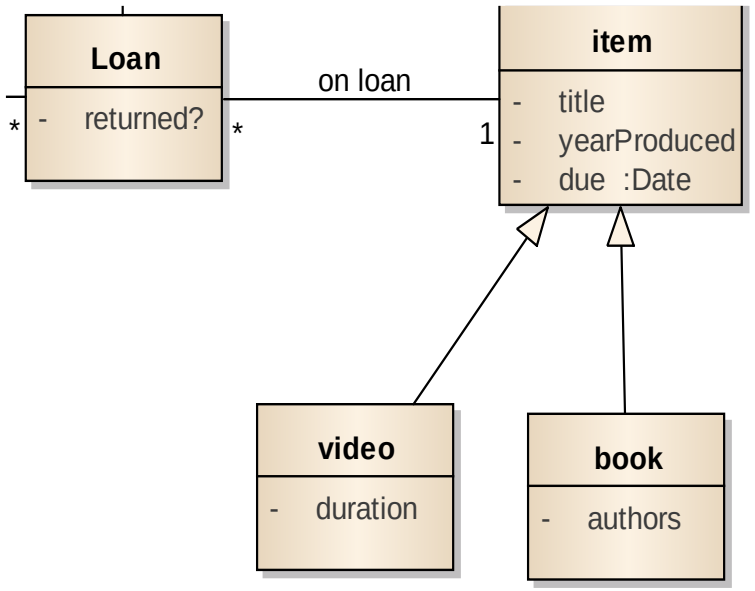
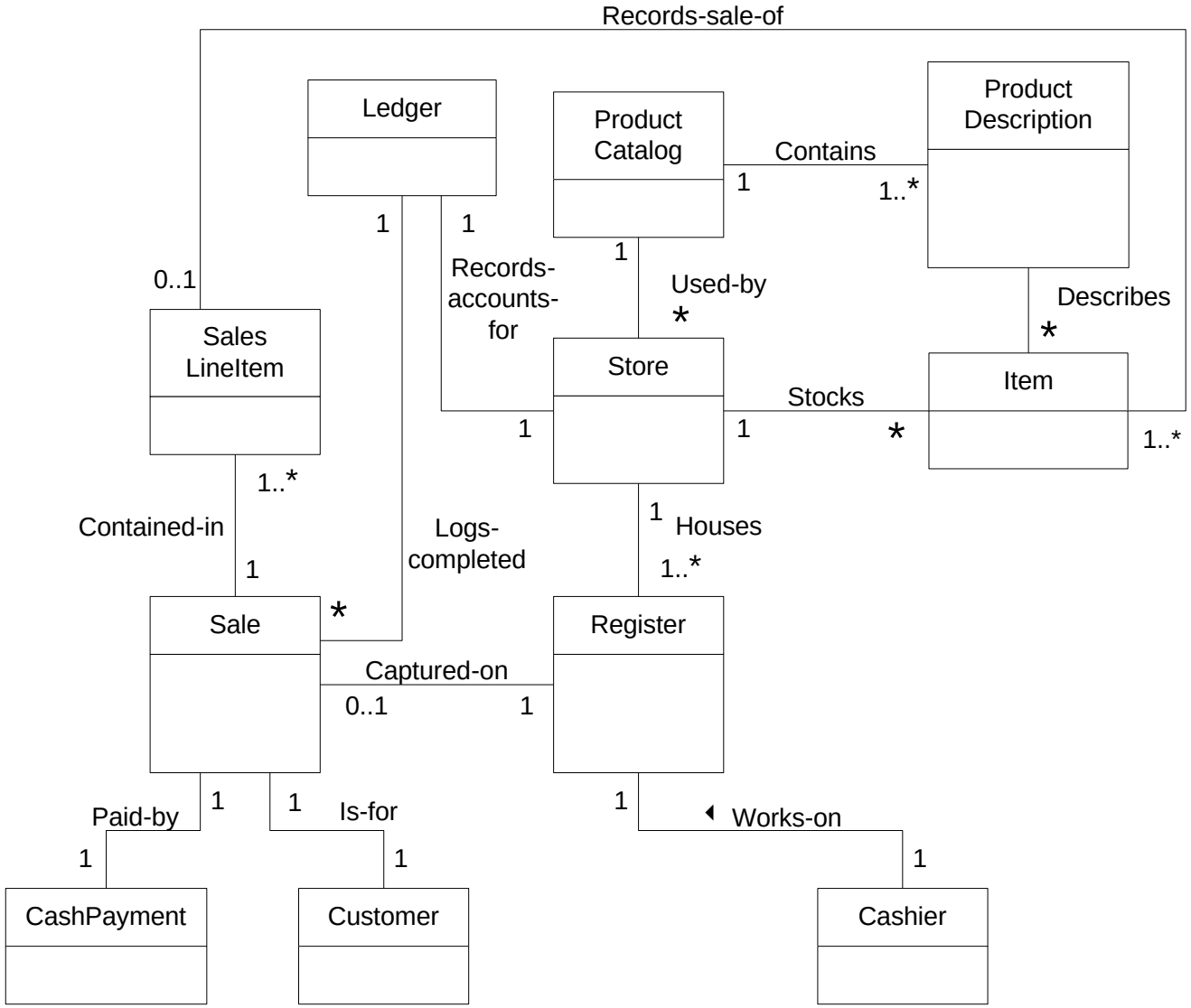


Partial domain model

POS: Partial Domain Model



Attributes of Conceptual Classes

- **Definition**: An *attribute* is a logical data value of an object.
- Attributes should be included into conceptual classes when the requirements (e.g., use cases) suggest or imply *a need to remember information*

Main Success Scenario (or Basic Flow):

1. Customer arrives at a POS checkout with goods and/or services to purchase.
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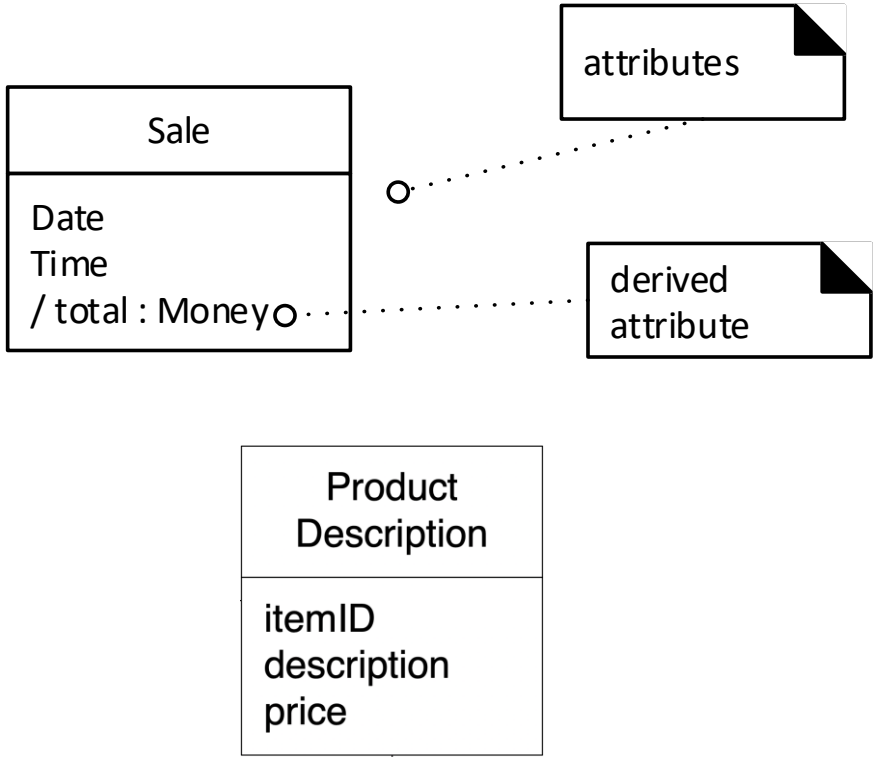
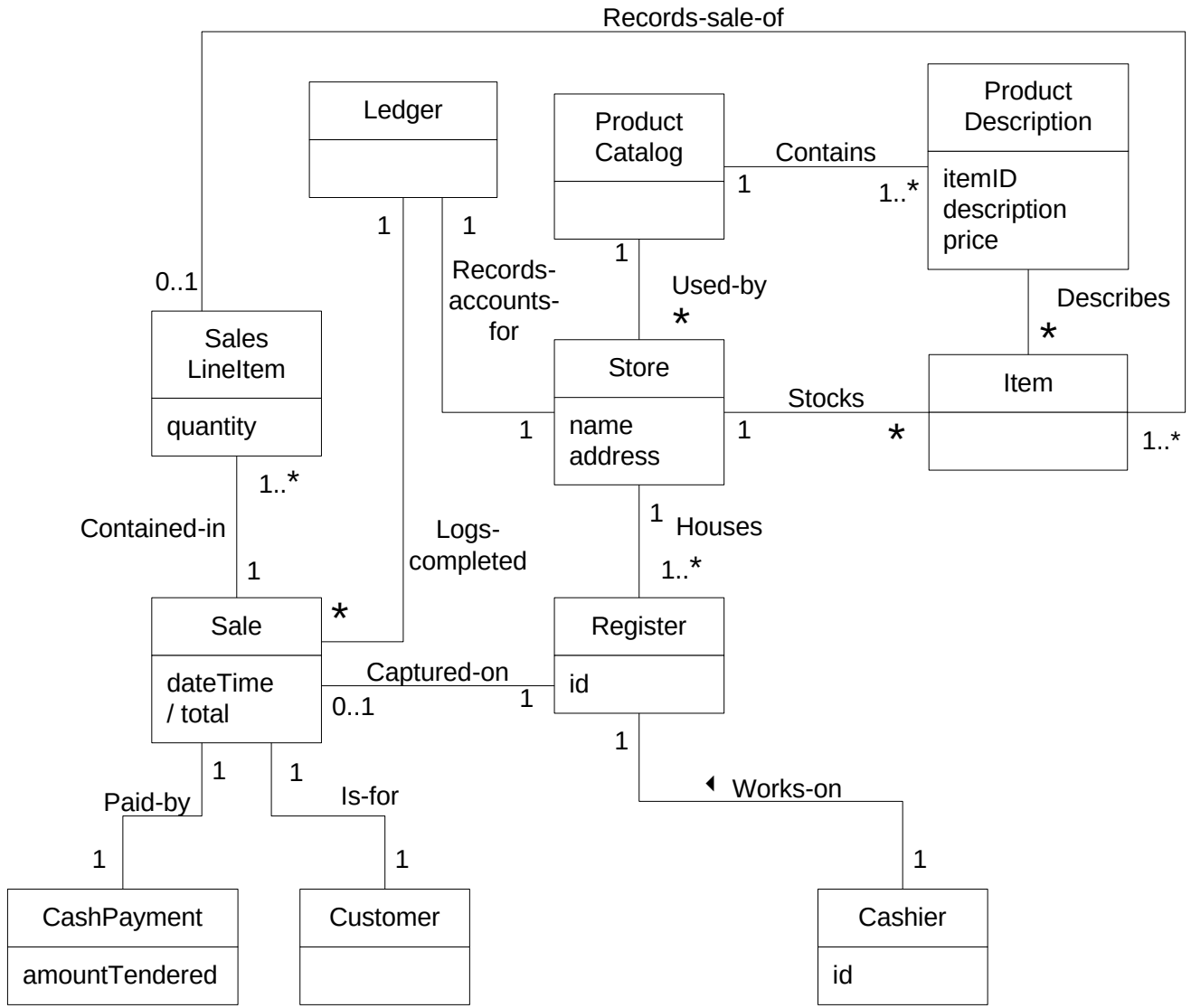


POS: Attributes in Model



Subclasses vs Attributes

The library has **users** who can borrow **items**.

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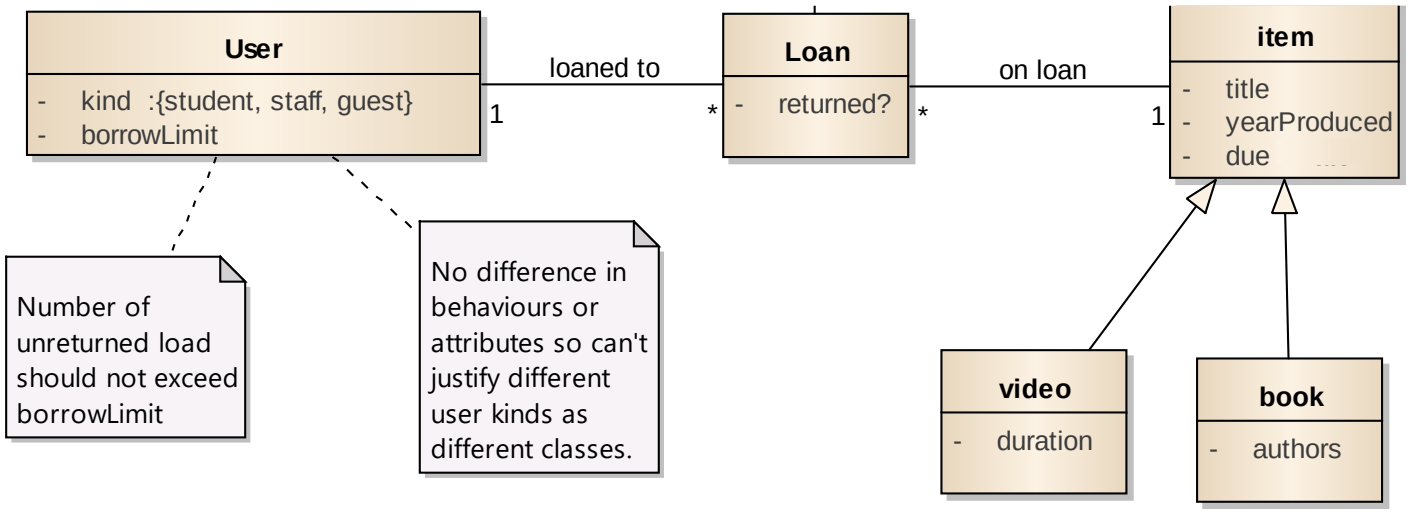

Partial domain model

Guideline for determining subclasses

Guideline for creating subclasses:

1. Subclass has additional attributes of interest
2. Subclass has additional associations of interest
3. Subclass is operated on, handled, reacted to, or manipulated differently to the other classes in noteworthy ways

Modelling Guidelines:

a) Declare superclasses abstract

b) Append the superclass name to the subclass

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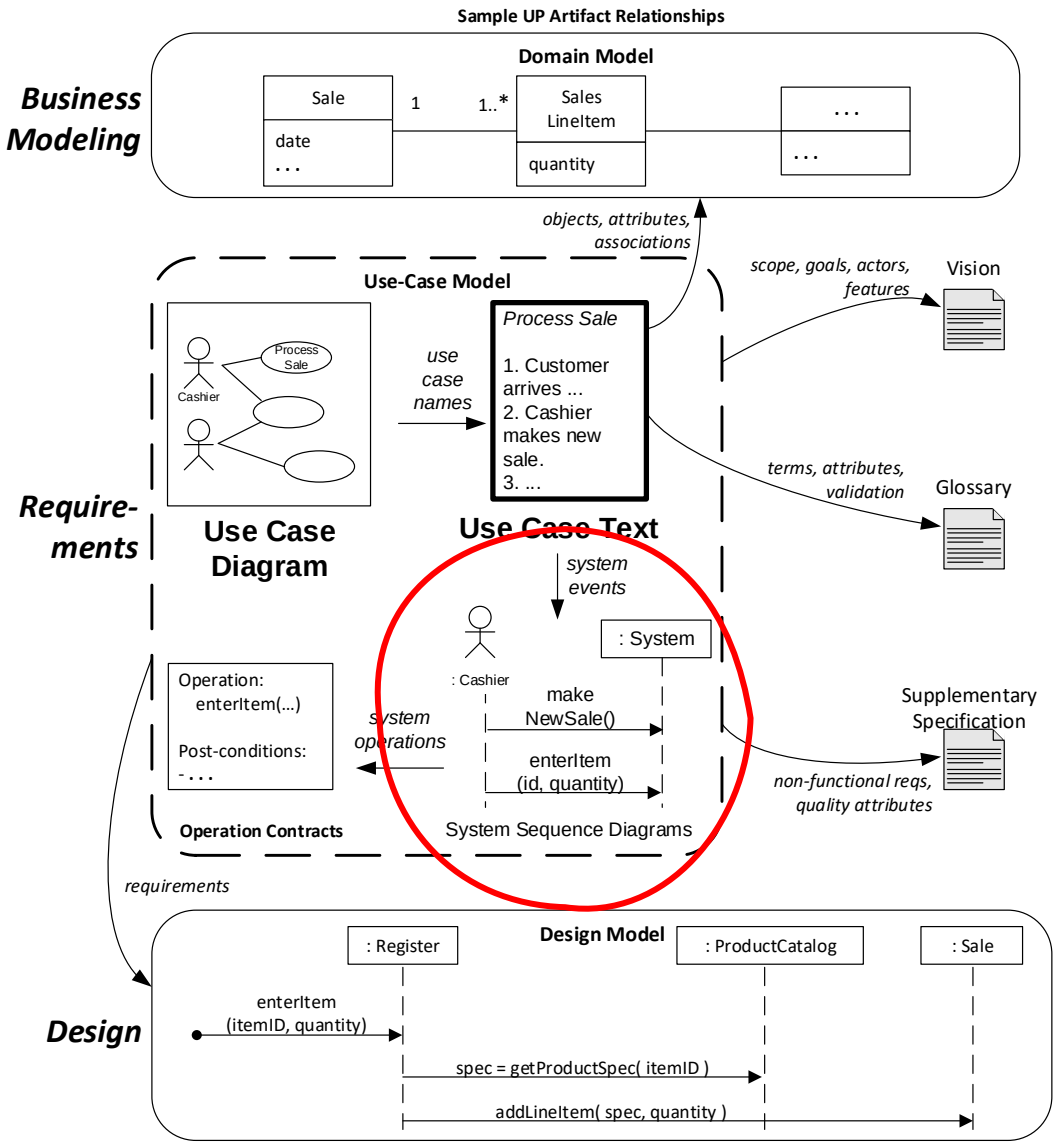
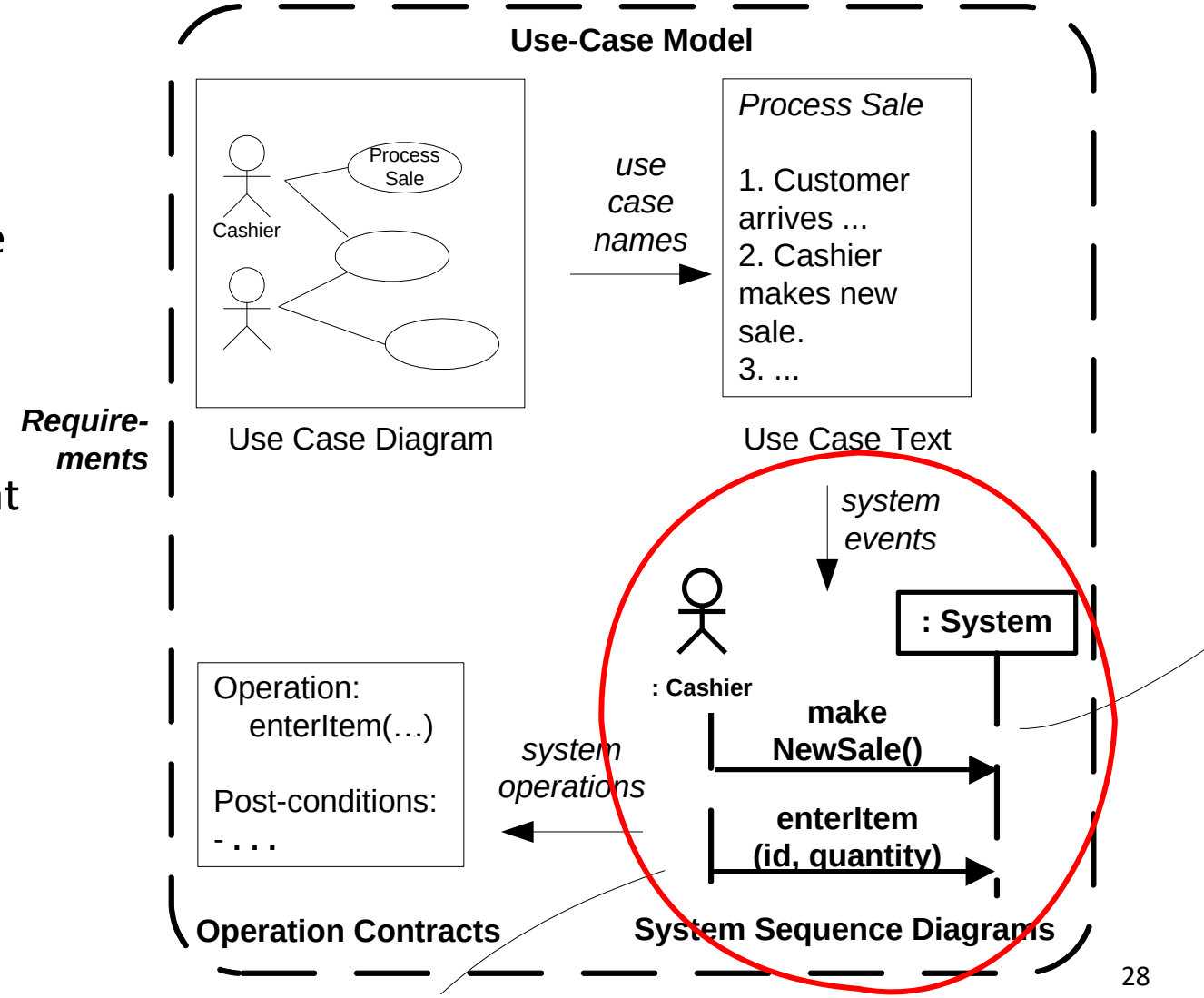
System Sequence Diagrams

Textbook: Larman Chapter 10



System Sequence Diagrams (SSDs)

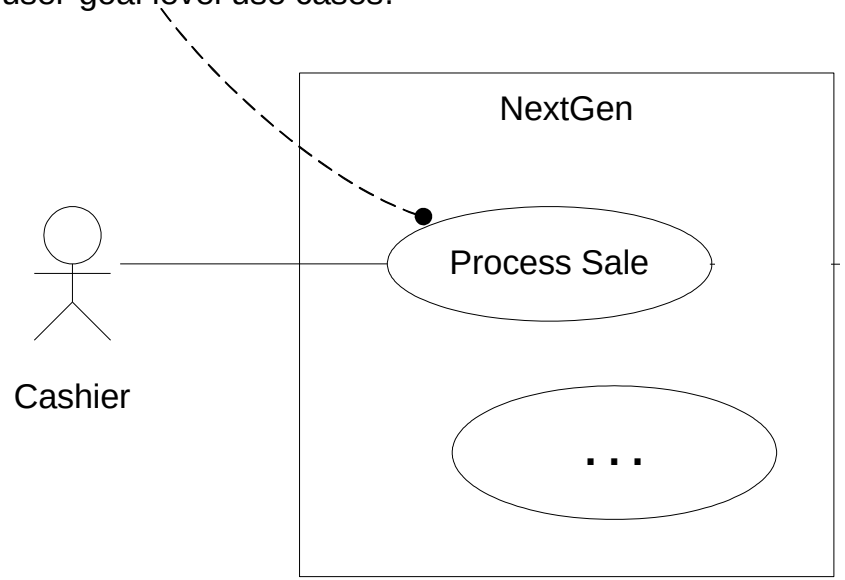
- **Definition:** A visualisation of the *system events* that external actors generate, the *order* of the events, and possible inter-system events
  - One SSD is for *one particular scenario* of a use case
  - Helps to identify the external input events to the system (the system events)
  - Treats system as a 'black box' to describe what the system does without explaining how it does it.



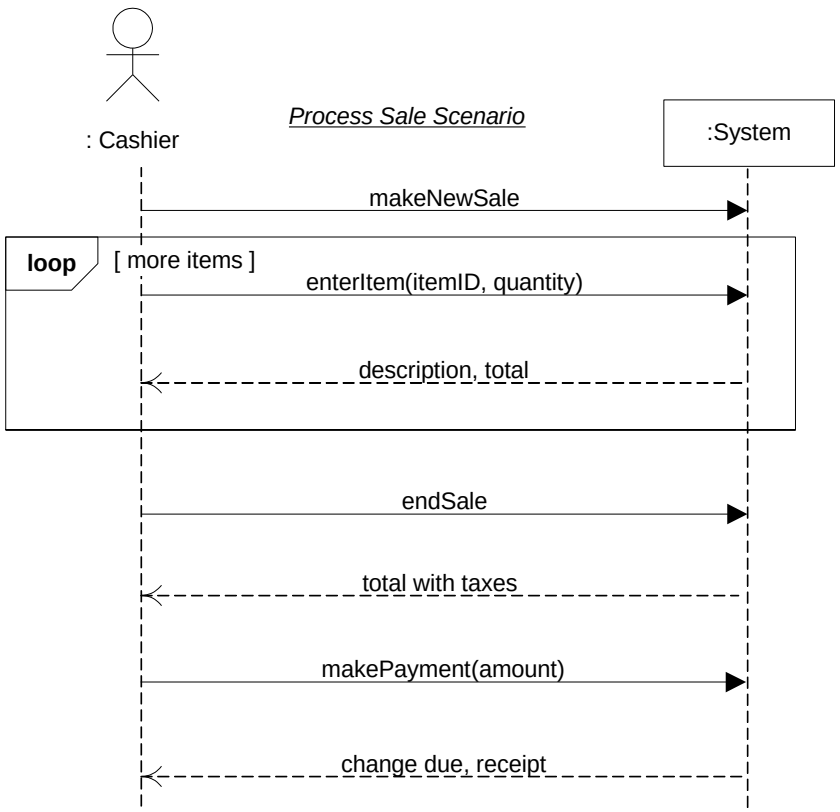
SSDs: Dynamic Context

- SSD captures dynamic context for system

For a use case context diagram, limit the use cases to user-goal level use cases.



Use Case Diagram: POS



SSD for POS Process Sale Scenario

SSDs: Dynamic Context

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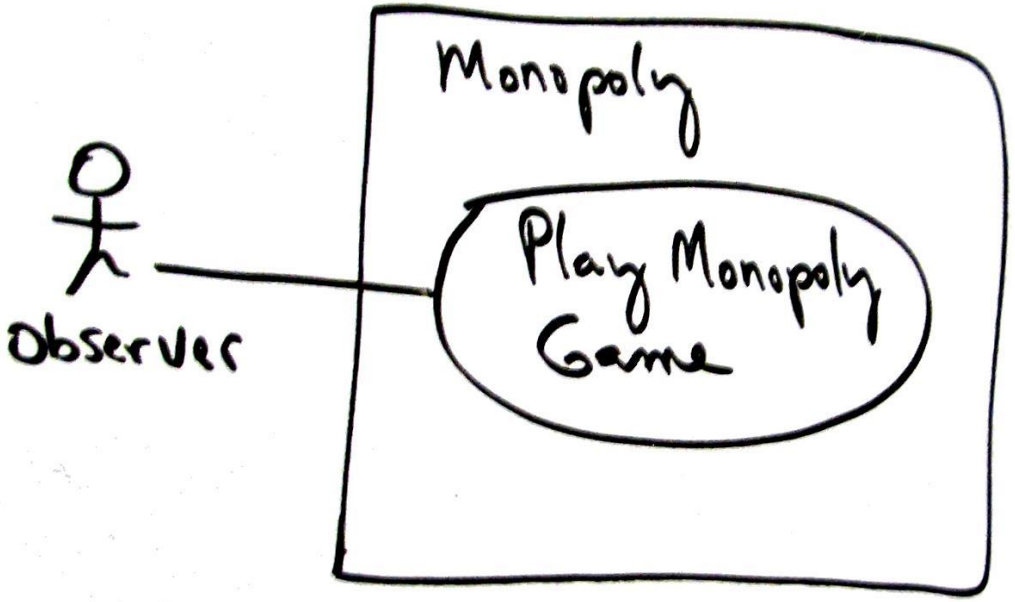
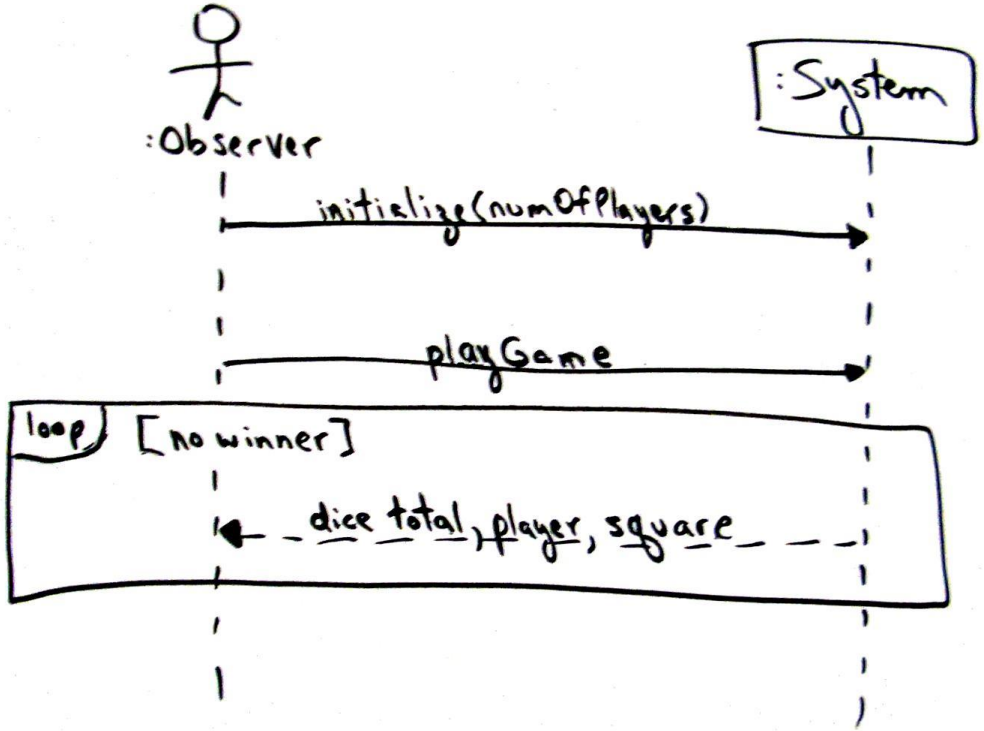


Use Case Diagram: Monopoly



SSD for a Play Monopoly Game Scenario

SSD: Process Sale scenario of POS use cases

Simple cash-only *Process Sale* scenario:

1. Customer arrives at a POS checkout with goods and/or services to purchase.
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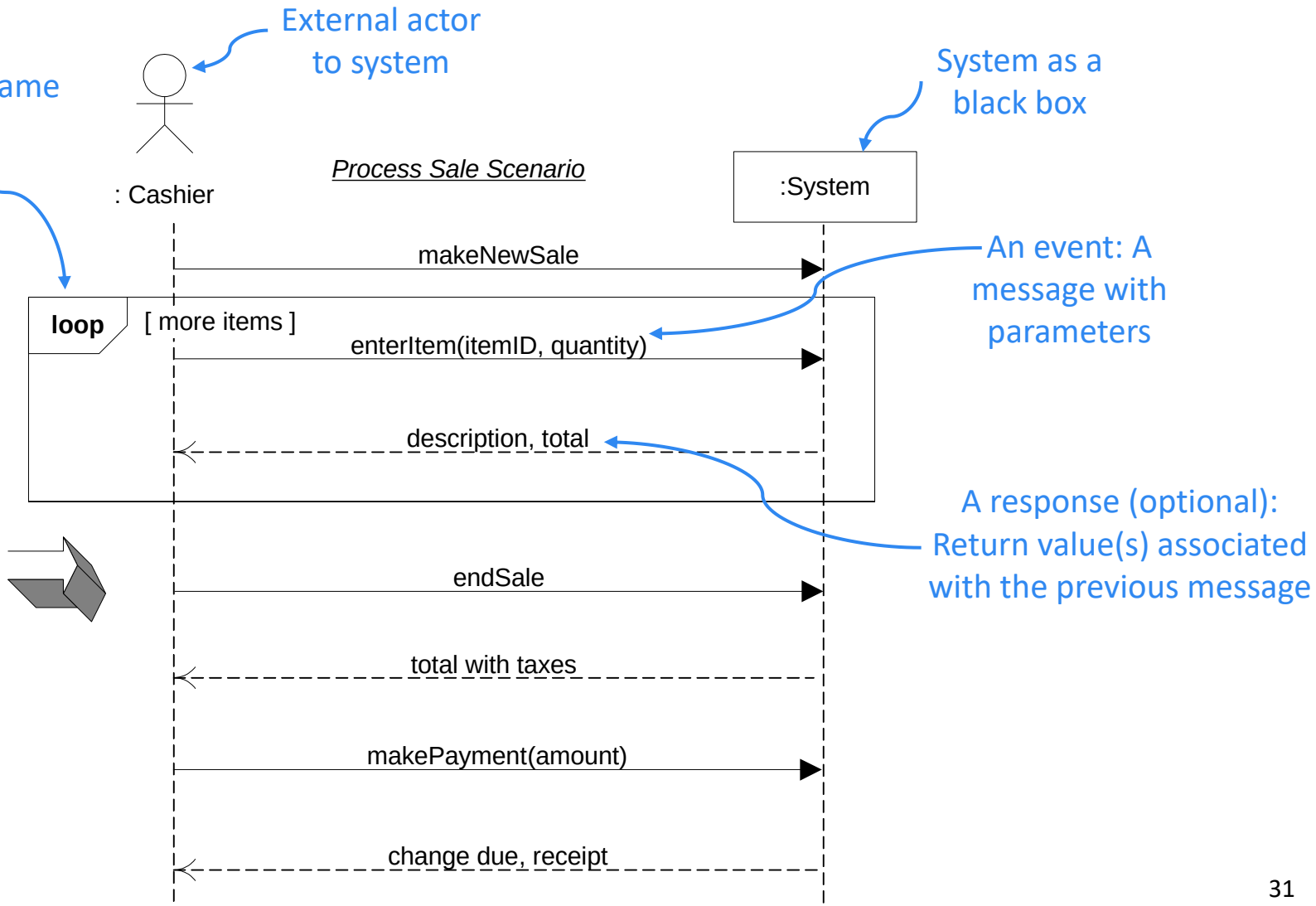
3. Cashier enters item identifier.
4. System records sale line item and presents item description, price, and running total.
Cashier repeats steps 3-4 until indicates done.
5. System presents total with taxes calculated.
6. Cashier tells Customer the total, and asks for payment.
7. Customer pays and System handles payment.
...



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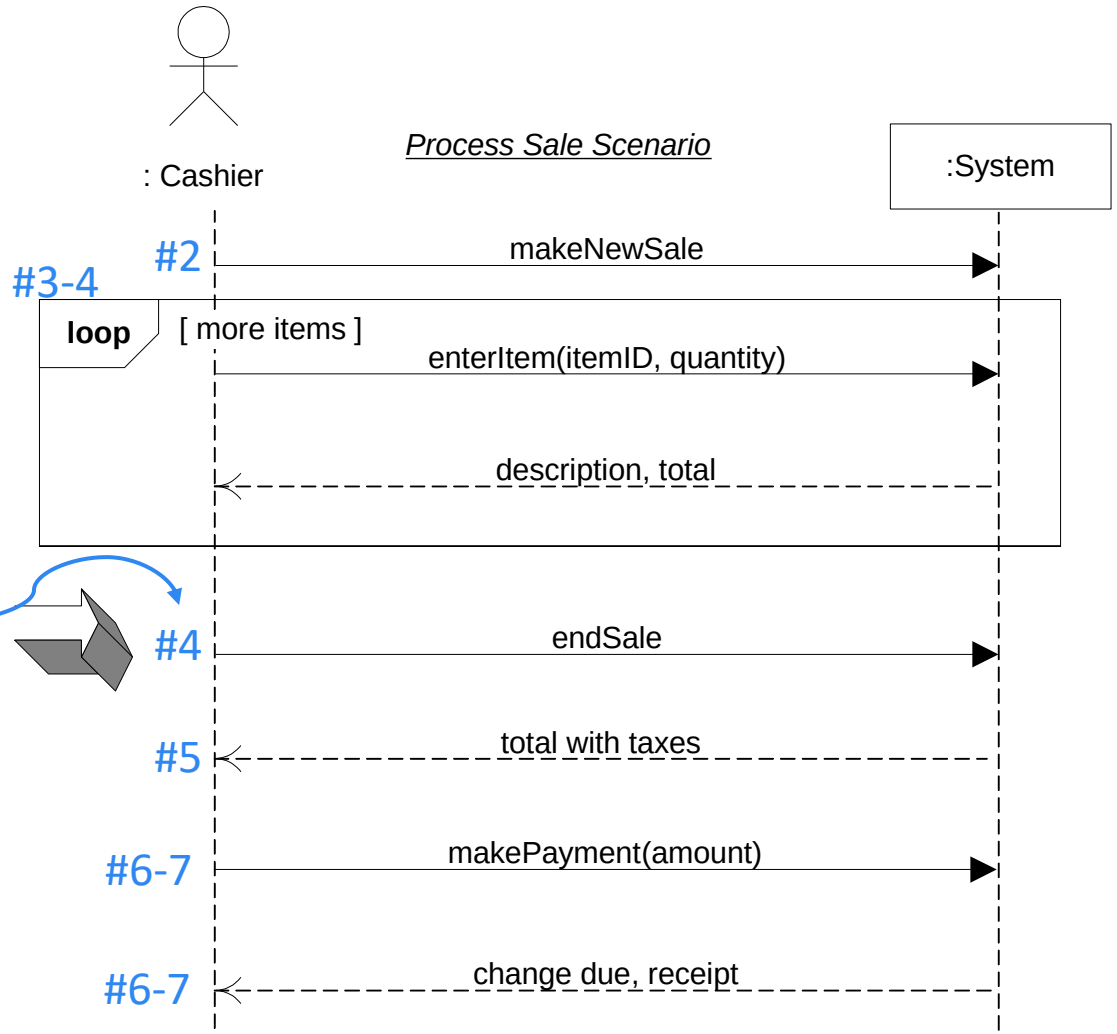

Exercise: SSD for Myki Quick Top-up Machine

Quick top-up scenario

1. A traveler enter a Myki card number by tapping the card on the green panel (the reader). The machine displays the balance of the card.
2. The traveler enter the amount of top up from the displayed menu.
3. Then, the traveler make a payment by tapping a credit card on the green panel.
4. A traveler then enter a Myki card number by tapping again to load the top up amount into the card. Finally, the machine displays the current balance of the card.

Myki: A smart card ticketing system

https://www.ptv.vic.gov.au/tickets/myki/travel-with-myki/myki-equipment/top-up-faster-with-a-quick-top-up-machine/

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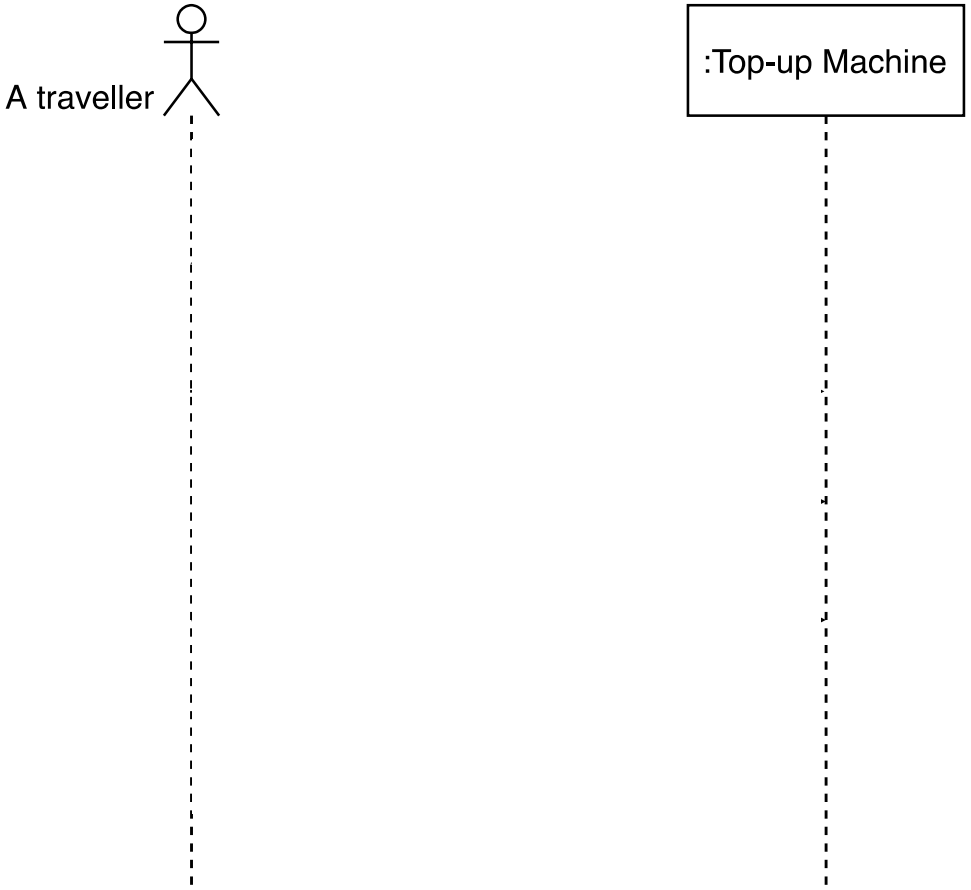

SSD for Myki Quick Top-up Machine

https://www.ptv.vic.gov.au/tickets/myki/travel-with-myki/myki-equipment/top-up-faster-with-a-quick-top-up-machine/

What is the FIRST system EVENT and RESPONSE (Return Message / Value) ?

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What is the SECOND system EVENT and RESPONSE (Return Message / Value) ?

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What is the THIRD system EVENT and RESPONSE (Return Message / Value) ?

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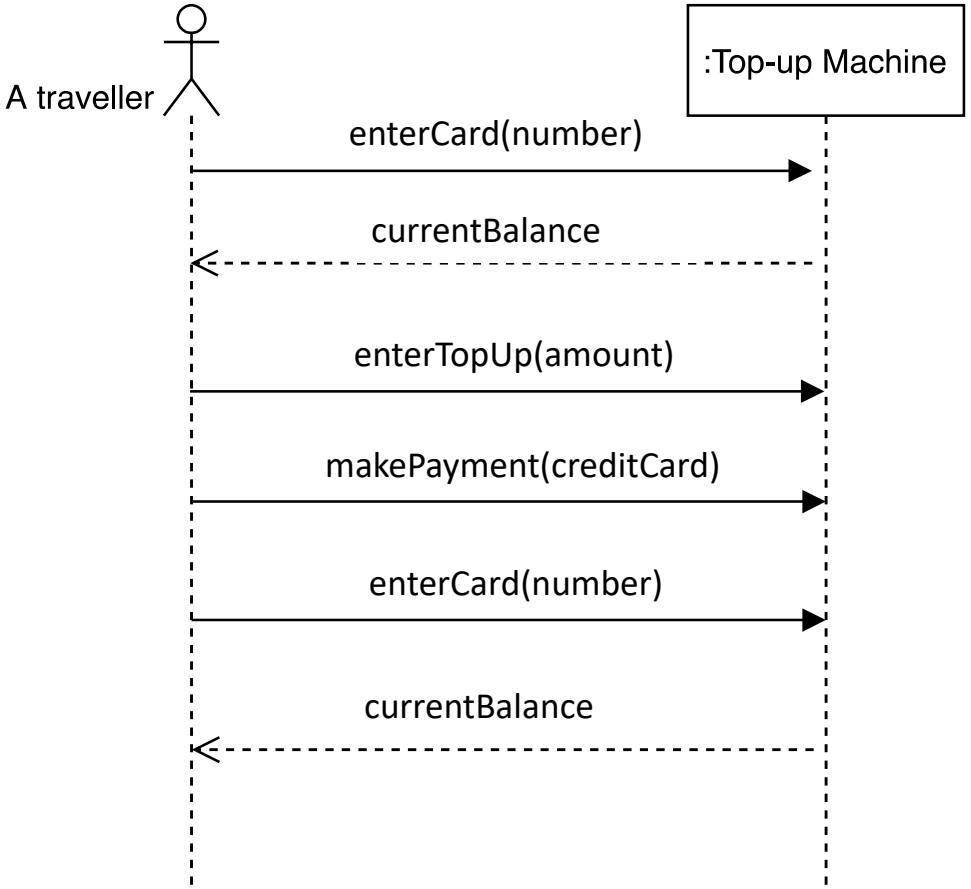

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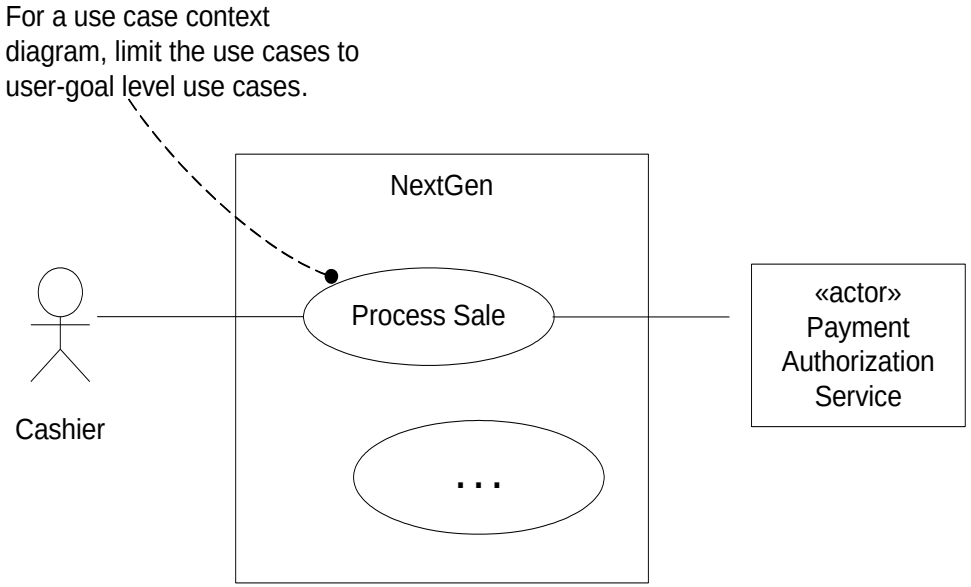
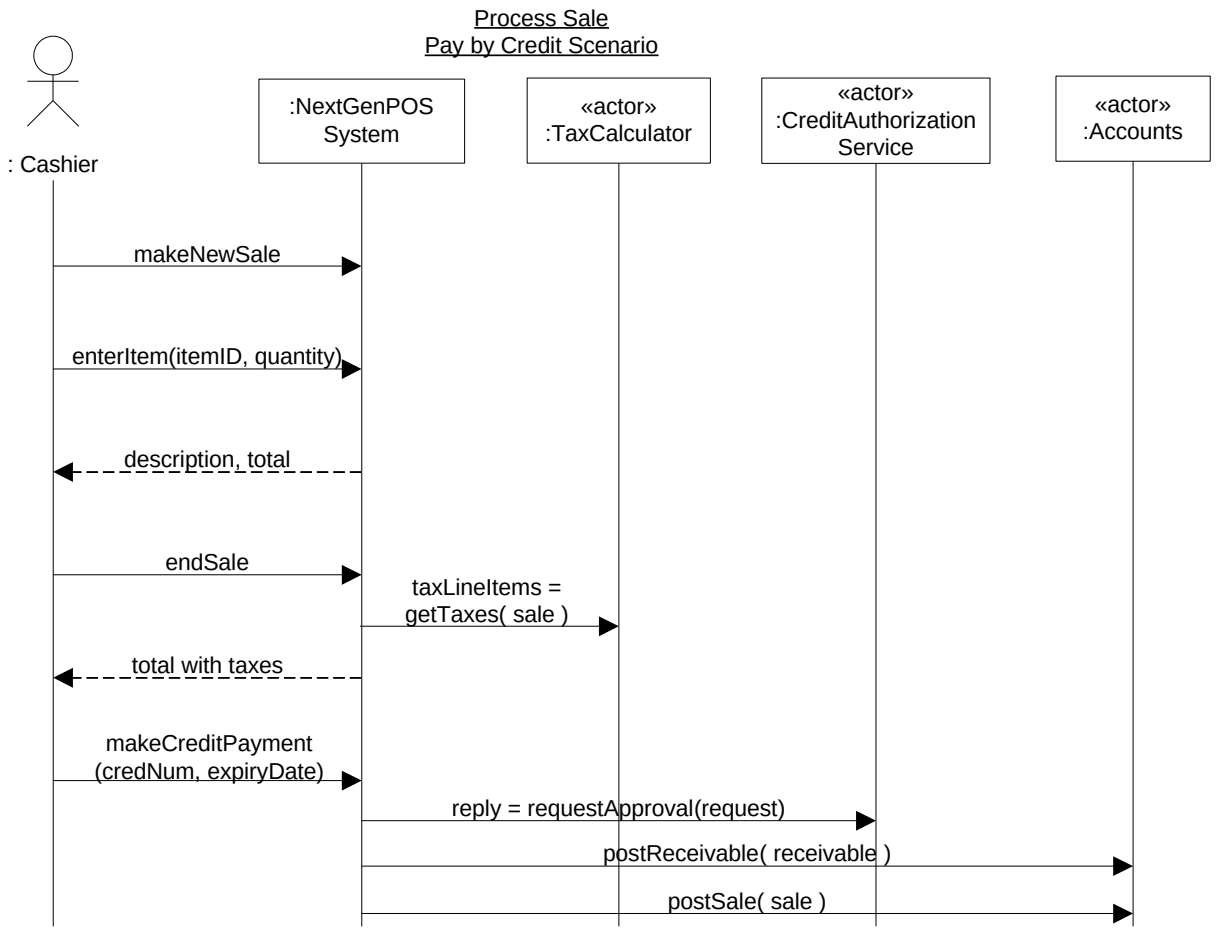
https://www.ptv.vic.gov.au/tickets/myki/travel-with-myki/myki-equipment/top-up-faster-with-a-quick-top-up-machine/

SSD: Inter-system events

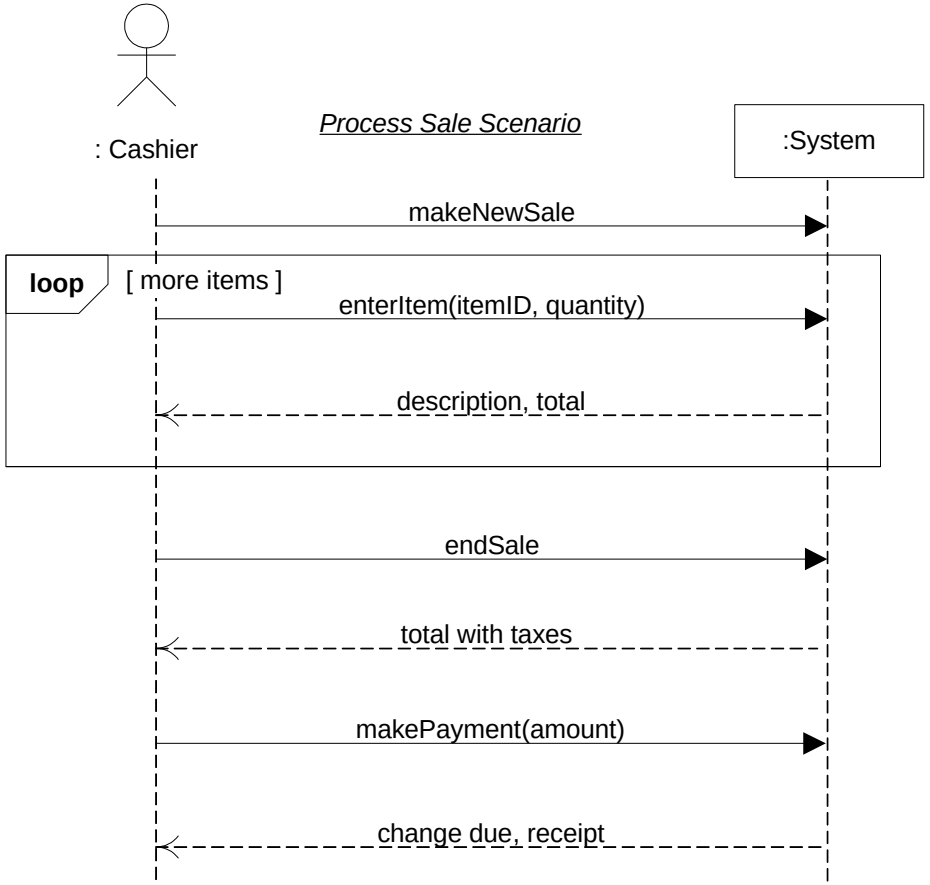
- SSDs can also illustrate interactions between systems, e.g., POS <--> External credit payment authorizer

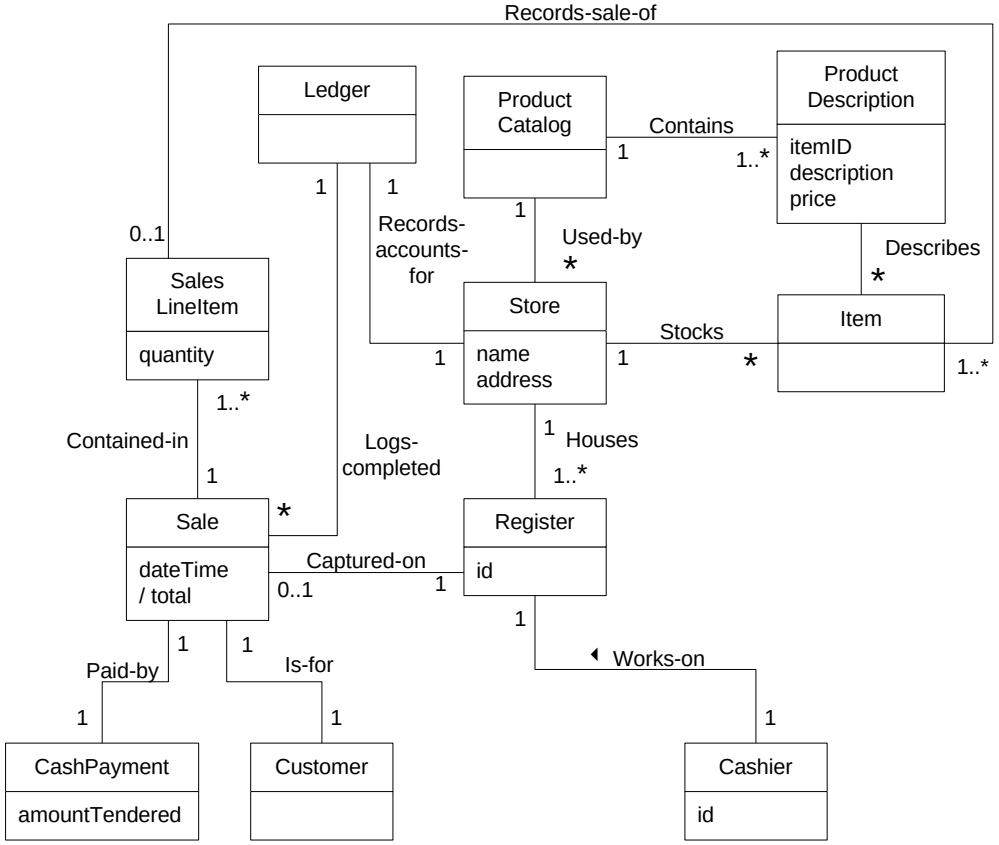




SSDs & Domain Models

- SSDs indicate events which design needs to handle and include associated information with reference to domain model





Summary & Remarks

- ***Object-oriented analysis*** concerns on how to describe the problem domain from the perspective of object
- **Domain models** capture the concepts, attributes, and associations in the domain
  - A visual dictionary of noteworthy abstractions, domain vocabulary, and information content of the domain
- **System Sequence Diagrams** capture the sequence of system events of one scenario in use cases
  - Provide *dynamic* context of system: How actors interact with system (as a black box) and how the system respond
- Domain models & SSDs should be expressed at the ***abstract level of intention***
  - Provide essential abstractions in information required to understand the domain in the context of the current requirements
  - Aid people in understanding the domain

SWEN30006 - Lecture Questions

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Nobody has responded yet.

Hang tight! Responses are coming in.

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Lecture Identification

Lecturer: Dr. Rajesh Chittor Sundaram

Semester 1, 2025

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These slides include materials from:

Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development, Third Edition, by Craig Larman, Pearson Education Inc., 2005.